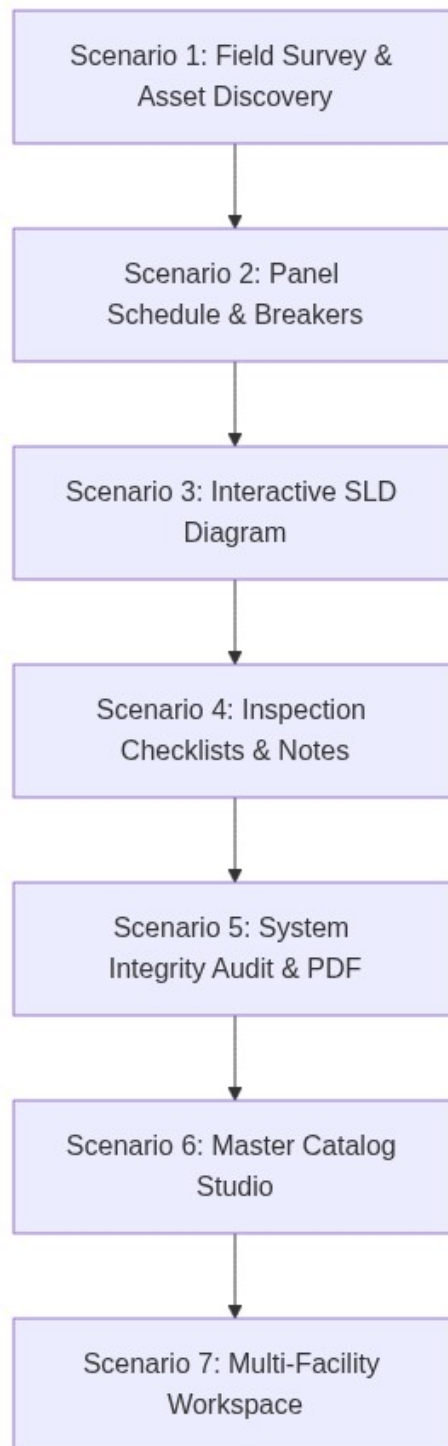


BuildingAware (JAMES) - Interactive Demo Scenarios & Script Guide

Executive Overview

BuildingAware (JAMES) is an intelligent electrical digital twin and field surveying platform designed for electrical engineers, facility managers, and safety auditors. It transforms physical electrical distribution systems into connected, code-validated digital models backed by SQLite, Git-friendly JSON Lines, and automated IEEE/NEC compliance engines.

This document outlines **7 high-impact demonstration scenarios** suitable for executive briefings, technical customer demos, and field surveyor training.



Scenario 1: Fast Field Walkthrough & Digital Twin Ingestion

Persona: Field Survey Technician / Electrical Auditor **Goal:** Demonstrate how fast and intuitive it is to survey physical equipment in a commercial or industrial facility on an iPad or laptop.

Demonstration Steps:

1. Navigate to Survey Workbench (/survey). 2. Archetype Selection:

- In **Column 1 (Domains)**, click ■ **Power Sources** or ■■ **Panelboards & MCC**.
- In **Column 2 (Equipment Types)**, click **Lighting Branch Panel (LP)**.
- *Talking Point: "Notice how JAMES automatically scans the facility stack and suggests the next sequential tag (LP-2 if LP-1 already exists), preventing tag duplication."*

3. Set Equipment Specs:

- Location: Select Level 1 - Electrical Closet 102.
- Voltage: 208Y/120V 3Ø 4W.
- Bus Rating: 225A (observe automatic NEC copper wire sizing: 4/0 AWG Cu).

4. Link Upstream Power Tree (Fed From):

- Open the **Fed From** dropdown.
- Select T-1 (75 kVA Step-Down Transformer).
- *Talking Point: "The smart hierarchy engine automatically filters out terminal loads and prevents cyclical circular references."*

5. Capture Photo Evidence & Notes:

- Tap ■ **Snap Photo / Upload** to attach a nameplate photo.
- Add a field observation: "Clean enclosure, mounted flush on north wall."

6. Save to Stack: Click ■ **Save Equipment**. Observe the asset immediately appearing in **Column 4 (Facility Stack)**.

Scenario 2: Interactive Panel Schedule, Breaker Ganging & Unresolved Loads

Persona: Senior Electrical Engineer / Project Estimator **Goal:** Showcase the NEMA-accurate odd/even 3-phase panel schedule and automated breaker linking.

Demonstration Steps:

1. Open MDP-1 (Main Distribution Switchboard, 42 Slots) in the equipment editor drawer. 2. Observe 3-Phase Color-Coded Rotation:

- Show the middle Ph column with anchored badges:
- ■ **Phase A** (Red)
- ■ **Phase B** (Blue)
- ● **Phase C** (Emerald)
- *Talking Point: "The alternating bus stabs accurately represent physical panel internals (Rows 1-2 = A, Rows 3-4 = B, Rows 5-6 = C)."*

3. Configure a 3-Pole Feeder Breaker:

- Click **Slot 1** (Feeder to Transformer T-1).
- Select **3-Pole** in the Breaker Inspector.
- Pick an OEM catalog model: Square D PowerPact H-Frame 100A 65kAIC.
- Observe how Slots 3 and 5 are automatically ganged across all three phases (Phase A, Phase B, Phase C).

4. Resolve Connected Loads with One Click:

- If a downstream panel (e.g. LP-1A) was added with Fed From: MDP-1, observe the amber ■■ Unresolved Connected Loads banner above the schedule.
 - Click **Resolve** ■.
 - The inspector automatically selects the downstream asset and finds the first available 3-pole slot pair.
 - Click **Save Breaker** to link the circuit.
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Scenario 3: Live Single-Line Diagram (SLD) Compilation

Persona: Consulting Engineer / Facility Architect **Goal:** Demonstrate real-time Graphviz WebAssembly diagram synthesis directly from the digital twin.

Demonstration Steps:

1. Click ■ **SLD Diagram** in the top navigation (/sld). 2. **Observe Live Rendering:**

- Point out the hierarchical tree flow:

UTIL-1 (Utility Grid) → ATS-1 (Transfer Switch) → MDP-1 (Main Switchboard) → T-1 (Transformer) → LP-1A (Branch Panel)

- Show standby generation branch: GEN-1 → ATS-1.

3. **Canvas Navigation:**

- Use the mouse wheel or pinch-to-zoom for smooth canvas navigation.
- Click **Fit to Screen** or toggle layout orientation (Top-to-Bottom vs Left-to-Right).

4. **Interactive Node Inspection:**

- Click any equipment box on the SVG diagram (e.g., MDP-1 or T-1).
- The slide-over editor drawer immediately opens with the full equipment record and panel schedule.
- Edit an attribute (e.g., change AIC from 65 kAIC to 100 kAIC) and save.
- The single-line diagram instantly re-compiles and updates in under 200 milliseconds.

5. **Export:** Click **Export SVG** or **Export DOT** for CAD / engineering deliverables.

Scenario 4: Field Inspection Checklists, Deficiency Tagging & Sign-Off

Persona: Safety Inspector / Commissioning Agent **Goal:** Demonstrate standards-compliant inspection workflows with one-tap defect chips and digital sign-off.

Demonstration Steps:

1. Open MDP-1 or LP-1A in the editor drawer and click the ■ **Checklists** tab. 2. **Domain-Specific Protocols:**

- Show how inspection checklists load based on equipment domain (e.g. *Visual Inspection, Working Space & NEC 110.26 Clearance, Arc Flash Labeling, Torque Verification*).

3. **1-Tap Status Lifecycle:**

- Click **Pass** (✓) on *Enclosure Integrity & Secure Mounting*.
- Click **Deficient** (■) on *NEC 110.26 Working Clearances*.

4. Auto-Revealed Deficiency Notes & One-Tap Preset Chips:

- Notice that marking an item as **Deficient** immediately expands the notes textarea with warning styling.
- Tap preset pills to instantly populate pithy field tags:
 - + Clearance < 36"
 - + Label Missing
 - + Torque Unverified
 - + Missing Filler Plate
 - + Arc Flash Expired

5. Filter & Live Compliance Metric:

- Toggle filters: All (14), Deficient (1), Pending (2), Passed (11).
- Observe the live score badge: 92% Compliant (1 Deficiency).

6. Inspector Sign-Off:

- Enter inspector name: Jane Doe, PE.
- Click **Sign Off Inspection**.
- A permanent timestamped verification badge is stamped onto the digital twin record.

Scenario 5: Automated System Integrity Audit & Publication-Ready PDF

Persona: Chief Electrical Engineer / Insurance Underwriter / AHJ **Goal:** Demonstrate automated NEC/IEEE engineering rule checks and server-side PDF report compilation.

Demonstration Steps:

1. Open the **Toolbox** (■) in the top header and click ■■ **Run Integrity Audit**. 2. **Deterministic Rule Engine Checks:**

- **Voltage Class Continuity:** Flags if a 480V panel directly feeds a 208V panel without a transformer.
- **Capacity & Headroom Analysis:** Verifies that upstream breakers have sufficient ampacity headroom for downstream load MCA/FLA.
- **AIC Fault Withstand:** Flags equipment with withstand ratings lower than available fault current.
- **Single Point of Failure (SPOF):** Highlights non-redundant critical distribution paths.

3. Review Audit Scorecard:

- Point out the **Facility Health Score** (e.g. 88 / 100) and categorized badges (Critical, Warning, Info).
- Click **Inspect Asset** ■ on a finding to jump straight to the offending equipment for correction.

4. Generate Publication-Grade PDF:

- Click the ■■ **Print / PDF** button.
- *Talking Point: "Rather than printing a messy browser screenshot, JAMES executes server-side typography to render a formal report with executive summary tables, health scorecards, itemized deficiency logs, and facility power trees."*

5. Alternative Exports: Click ■ **Export Markdown (TOC)** or ■ **Export CSV** for spreadsheet integration.

Scenario 6: Master Vendor Catalog Studio & Cut-Sheet PDF Viewer

Persona: Procurement Specialist / Electrical Estimator **Goal:** Show the verified 5,000+ manufacturer component catalog and integrated technical documentation.

Demonstration Steps:

1. Navigate to ■ **Catalog Studio** (/catalog). 2. **Filter by Manufacturer & Archetype:**

- Select Manufacturer: Square D / Schneider Electric.
- Select Archetype: Molded Case Circuit Breaker (MCCB).
- Filter by Pole Count: 3-Pole, Trip Rating: 100A - 400A.

3. **Inspect Catalog Item:**

- View real-world Frame sizes, Series (e.g., PowerPact H / J), AIC ratings (25kA, 65kA, 100kA), and mounting styles (Bolt-on, I-Line).

4. **Embedded PDF Technical Cut-Sheets:**

- Click **View Cut-Sheet** ■ on any indexed part.
 - The integrated PDF viewer loads the manufacturer's technical document with zoom, page navigation, and search highlights.
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Scenario 7: Multi-Facility Workspaces, Offline-First & Knowledge Base

Persona: Enterprise Operations Lead / Fleet Manager **Goal:** Showcase enterprise client isolation, offline resilience, and searchable engineering standards.

Demonstration Steps:

1. **Facility Database Switching:**

- Press Cmd + 0 (or open Toolbox ■ → **Facility Finder**).
- Show multiple client workspaces (e.g. zoetis / b4, zoetis / b17_test, pfizer / plant_1).
- Switch between facilities instantly without page reloads.

2. **Search the Integrated Knowledge Base:**

- In the Toolbox Help & FAQ search bar, type: "arc flash labeling" or "working clearance".
- Show deep-linked search results pointing directly to indexed PDF standards (e.g. QSG-Arc Flash Labeling.pdf, Page 1).

3. **Display Scaling & Low-Light Theme:**

- Toggle ■ **Dark Mode** for dimly lit electrical basements and ■■ **Light Mode** for outdoor substations.
- Adjust Font & UI Scaling (Compact, Standard, Large Touch) for tablet field ergonomics.

4. **Dual Persistence Verification:**

- Open the data folder to show atomic SQLite (model.db) operating in tandem with Git-friendly JSON Lines (model.jsonl).
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Quick Reference Demo Cheat Sheet

Feature	Where to Click	Key Talking Point
Survey Discovery	/survey → Left Columns	10 standard digital twin domains; auto-incrementing tags
Panel Schedule	Survey / SLD Drawer → Schedule	NEMA odd/even 3-phase layout with color-coded A/B/C badges
SLD Diagram	/sld in Top Header	Real-time Graphviz WebAssembly rendering in <200ms
Checklists	Equipment Drawer → Checklists Tab	1-tap deficiency status + preset defect pill chips
Integrity Audit	Header ■ → Run Integrity Audit	Deterministic NEC/IEEE rule validation + Health Score
PDF Report	Audit Modal → ■■ Print / PDF	Publication-grade server-side formatted engineering audit report
Catalog Studio	/catalog in Top Header	5,000+ verified OEM parts with integrated PDF cut-sheets
Facility Finder	Header Pill or Cmd + O	Instant multi-client workspace switching with zero reload

Prepared by BuildingAware Engineering Team — Powered by JAMES Digital Twin Platform.